FAITHFULNESS OF THE MODULAR-GROUP ACTION ON THE UNIVERSAL ASKEY–WILSON ALGEBRA

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Allowed standard prerequisites (editorial).

- PSL2(Z)=C3*C2 with reduced alternating-word normal form. Alperin1993, original bibliography alpern. Representation construction and nontrivial reduced-word test.
- Center of Mat2(F[lambda,lambda^-1]) consists of scalar matrices. Standard ring-theoretic fact; author statement and short proof marker retained at628–642.
- Basic polynomial degree, determinant and matrix multiplication. Leading coefficient and determinant contradiction.

Our conventions for the paper are as follows. An algebra is meant to be associative and have a 1. A subalgebra has the same 1 as the parent algebra. We fix a field $\mathbb{F}$ and a nonzero $q \in \mathbb{F}$ such that $q^4 \neq 1$.

Definition 0.1. Define an $\mathbb{F}$-algebra $\Delta = \Delta_q$ by generators and relations in the following way. The generators are A, B, C. The relations assert that each of

$$A + \frac{qBC - q^{-1}CB}{q^2 - q^{-2}}, \qquad B + \frac{qCA - q^{-1}AC}{q^2 - q^{-2}}, \qquad C + \frac{qAB - q^{-1}BA}{q^2 - q^{-2}} \tag{0.1}$$

is central in Δ. We call Δ the *universal Askey–Wilson algebra*.

Definition 0.2. For the three central elements in (0.1), multiply each by $q + q^{-1}$ to get α, β, γ. Thus

$$A + \frac{qBC - q^{-1}CB}{q^2 - q^{-2}} = \frac{\alpha}{q + q^{-1}}, \tag{0.2}$$

$$B + \frac{qCA - q^{-1}AC}{q^2 - q^{-2}} = \frac{\beta}{q + q^{-1}}, \tag{0.3}$$

$$C + \frac{qAB - q^{-1}BA}{q^2 - q^{-2}} = \frac{\gamma}{q + q^{-1}}. \tag{0.4}$$

Note that each of α, β, γ is central in Δ. (The purpose of the factor $q + q^{-1}$ is to make the upcoming formula (0.5) more attractive.)

$$\Omega = qABC + q^2A^2 + q^{-2}B^2 + q^2C^2 - qA\alpha - q^{-1}B\beta - qC\gamma. \tag{0.5}$$

Lemma 0.3. *The $\mathbb{F}$-algebra Δ is generated by A, B, γ. Moreover*

$$C = \frac{\gamma}{q + q^{-1}} - \frac{qAB - q^{-1}BA}{q^2 - q^{-2}}, \tag{0.6}$$

$$\alpha = \frac{B^2A - (q^2 + q^{-2})BAB + AB^2 + (q^2 - q^{-2})^2A + (q - q^{-1})^2B\gamma}{(q - q^{-1})(q^2 - q^{-2})}, \tag{0.7}$$

$$\beta = \frac{A^2B - (q^2 + q^{-2})ABA + BA^2 + (q^2 - q^{-2})^2B + (q - q^{-1})^2A\gamma}{(q - q^{-1})(q^2 - q^{-2})}. \tag{0.8}$$

Proof. Line (0.6) is from (0.4). To get (0.7), (0.8) eliminate C in (0.2), (0.3) using (0.6). □

Recall the notation

$$[n]_q = \frac{q^n - q^{-n}}{q - q^{-1}}, \qquad n = 0, 1, 2, \ldots.$$

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Theorem 0.4. *The $\mathbb{F}$-algebra Δ has a presentation by generators A, B, γ and relations*

$$A^3B - [3]_q A^2BA + [3]_q ABA^2 - BA^3 = -\big(q^2 - q^{-2}\big)^2 (AB - BA),$$
$$B^3A - [3]_q B^2AB + [3]_q BAB^2 - AB^3 = -\big(q^2 - q^{-2}\big)^2 (BA - AB),$$
$$A^2B^2 - B^2A^2 + \big(q^2 + q^{-2}\big)(BABA - ABAB) = -\big(q - q^{-1}\big)^2 (AB - BA)\gamma,$$
$$\gamma A = A\gamma, \qquad \gamma B = B\gamma.$$

Proof. Use Lemma 0.3 to express the defining relations for Δ in terms of A, B, γ. □

We now consider some automorphisms of Δ. Recall that the modular group $\mathrm{PSL}_2(\mathbb{Z})$ has a presentation by generators ρ, σ and relations $\rho^3 = 1$, $\sigma^2 = 1$. See for example [1].

Theorem 0.5. *The group $\mathrm{PSL}_2(\mathbb{Z})$ acts on Δ as a group of automorphisms in the following way:*

u	A	B	C	α	β	γ
$\rho(u)$	B	C	A	β	γ	α
$\sigma(u)$	B	A	$C + \frac{AB-BA}{q-q^{-1}}$	β	α	γ

Proof. By Definition 0.1 there exists an automorphism P of Δ that sends

$$A \mapsto B, \qquad B \mapsto C, \qquad C \mapsto A.$$

Observe $P^3 = 1$. By (0.2)–(0.4) the map P sends

$$\alpha \mapsto \beta, \qquad \beta \mapsto \gamma, \qquad \gamma \mapsto \alpha.$$

By Theorem 0.4 there exists an automorphism S of Δ that sends

$$A \mapsto B, \qquad B \mapsto A, \qquad \gamma \mapsto \gamma.$$

Observe $S^2 = 1$. By Lemma 0.3 the map S sends

$$\alpha \mapsto \beta, \qquad \beta \mapsto \alpha, \qquad C \mapsto C + \frac{AB - BA}{q - q^{-1}}.$$

The result follows. □

In Theorem 0.5 we gave an action of $\mathrm{PSL}_2(\mathbb{Z})$ on Δ. Our next goal is to show that this action is faithful.

Let λ denote an indeterminate. Let $\mathbb{F}[\lambda, \lambda^{-1}]$ denote the $\mathbb{F}$-algebra consisting of the Laurent polynomials in λ that have all coefficients in $\mathbb{F}$. We will be discussing the $\mathbb{F}$-algebra

$$\Lambda = \mathrm{Mat}_2(\mathbb{F}) \otimes_{\mathbb{F}} \mathbb{F}\big[\lambda, \lambda^{-1}\big].$$

We view elements of Λ as 2×2 matrices that have entries in $\mathbb{F}[\lambda, \lambda^{-1}]$. From this point of view the product operation for Λ is ordinary matrix multiplication, and the multiplicative identity in Λ is the identity matrix I. For notational convenience define $\mu = \lambda + \lambda^{-1}$.

For later use we now describe the center $Z(\Lambda)$.

Lemma 0.6. *For all $\eta \in \Lambda$ the following (i), (ii) are equivalent:*

(i) $\eta \in Z(\Lambda)$.
(ii) *There exists $\theta \in \mathbb{F}[\lambda, \lambda^{-1}]$ such that $\eta = \theta I$.*

Proof. Routine. □

Definition 0.7. Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ denote the following elements of Λ:

$$\mathcal{A} = \begin{pmatrix} \lambda & 1 - \lambda^{-1} \\ 0 & \lambda^{-1} \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} \lambda^{-1} & 0 \\ \lambda - 1 & \lambda \end{pmatrix}, \qquad \mathcal{C} = \begin{pmatrix} 1 & 1 - \lambda \\ \lambda^{-1} - 1 & \lambda + \lambda^{-1} - 1 \end{pmatrix}.$$

Lemma 0.8. *We have*

$$\mathcal{ABC} = I, \qquad \mathcal{A} + \mathcal{A}^{-1} = \mu I, \qquad \mathcal{B} + \mathcal{B}^{-1} = \mu I, \qquad \mathcal{C} + \mathcal{C}^{-1} = \mu I.$$

Proof. Use Definition 0.7. □

Lemma 0.9. *In the algebra Λ the elements $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ multiply as follows:*

	$\mathcal{A}$	$\mathcal{B}$	$\mathcal{C}$
$\mathcal{A}$	$\mu\mathcal{A} - I$	$\mu I - \mathcal{C}$	$\mu\mathcal{A} + \mathcal{B} + \mu\mathcal{C} - \mu^2 I$
$\mathcal{B}$	$\mu\mathcal{B} + \mathcal{C} + \mu\mathcal{A} - \mu^2 I$	$\mu\mathcal{B} - I$	$\mu I - \mathcal{A}$
$\mathcal{C}$	$\mu I - \mathcal{B}$	$\mu\mathcal{C} + \mathcal{A} + \mu\mathcal{B} - \mu^2 I$	$\mu\mathcal{C} - I$

Proof. Use Lemma 0.8. □

The algebra Λ is not generated by $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$. However we do have the following result.

Lemma 0.10. *Suppose $\eta \in \Lambda$ commutes with at least two of $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$. Then $\eta \in Z(\Lambda)$.*

Proof. For $1 \leq i, j \leq 2$ let η_{ij} denote the (i,j)-entry of η. The matrix η commutes with each of $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ since $\mathcal{ABC} = I$. In the equation $\eta\mathcal{A} = \mathcal{A}\eta$, evaluate $\mathcal{A}$ using Definition 0.7, and simplify the result to get $\eta_{21} = 0$. Similarly using $\eta\mathcal{B} = \mathcal{B}\eta$ we find $\eta_{12} = 0$ and $\eta_{11} = \eta_{22}$. Therefore $\eta = \eta_{11} I \in Z(\Lambda)$. □

Next we describe an action of $\mathrm{PSL}_2(\mathbb{Z})$ on Λ as a group of automorphisms.

Definition 0.11. Let p and s denote the following elements of Λ:

$$p = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}, \qquad s = \begin{pmatrix} 0 & 1 \\ \lambda & 0 \end{pmatrix}.$$

Lemma 0.12. *The following (i)–(iv) hold.*

(i) $\det(p) = 1$ *and* $\det(s) = -\lambda$.
(ii) $p^3 = -I$ *and* $s^2 = \lambda I$.
(iii) $p\mathcal{A}p^{-1} = \mathcal{B}$, $p\mathcal{B}p^{-1} = \mathcal{C}$, $p\mathcal{C}p^{-1} = \mathcal{A}$.
(iv) $s\mathcal{A}s^{-1} = \mathcal{B}$ *and* $s\mathcal{B}s^{-1} = \mathcal{A}$.

Proof. (i), (ii) Use Definition 0.11. (iii), (iv) Use Definition 0.7 and Definition 0.11. □

Lemma 0.13. *The group $\mathrm{PSL}_2(\mathbb{Z})$ acts on Λ as a group of automorphisms such that $\rho(\eta) = p\eta p^{-1}$ and $\sigma(\eta) = s\eta s^{-1}$ for all $\eta \in \Lambda$.*

Proof. By Lemma 0.12(ii) the elements p^3, s^2 are in $Z(\Lambda)$. □

Lemma 0.14. *The action of $\mathrm{PSL}_2(\mathbb{Z})$ on Λ is faithful.*

Proof. Pick an integer $n \geq 1$. Consider an element $\eta \in \Lambda$ of the form $\eta = \eta_1\eta_2\cdots\eta_n$ such that for $1 \leq i \leq n$, $\eta_i = s$ for one parity of i and $\eta_i \in \{p, p^{-1}\}$ for the other parity of i. We show that $\eta \notin Z(\Lambda)$. To this end we assume $\eta \in Z(\Lambda)$ and get a contradiction. Let ℓ denote the number of times s occurs among $\{\eta_i\}_{i=1}^n$. Assume for the moment $\ell = 0$. Then $n = 1$ so $\eta = \eta_1 \in \{p, p^{-1}\}$. The elements p, p^{-1} are not in $Z(\Lambda)$, for a contradiction. Therefore $\ell \neq 0$. From the nature of the matrices p, s in Definition 0.11, we may view η as a polynomial in λ that has coefficients in $\mathrm{Mat}_2(\mathbb{F})$ and degree at most ℓ. Call this polynomial f. We claim that the degree of f is exactly ℓ. To prove the claim, write

$$s = s_0 + s_1\lambda, \qquad s_0 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad s_1 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Let $m \in \mathrm{Mat}_2(\mathbb{F})$ denote the coefficient of λ^ℓ in f. The matrix m is obtained from $\eta_1\eta_2\cdots\eta_n$ by replacing each occurrence of s by s_1. Using $s_1ps_1 = -s_1$ and $s_1p^{-1}s_1 = s_1$ we find $m \in \{\pm p^i s_1 p^j | -1 \leq i, j \leq 1\}$. The matrix p is invertible and $s_1 \neq 0$ so $p^i s_1 p^j \neq 0$ for $-1 \leq i, j \leq 1$. Therefore $m \neq 0$ and the claim is proved. Let κ denote the $(1,1)$-entry of the matrix η. Then $\eta = \kappa I$ since $\eta \in Z(\Lambda)$. In the equation $\eta_1\eta_2\cdots\eta_n = \kappa I$ take the determinant of each side and use Lemma 0.12(i) to get $(-\lambda)^\ell = \kappa^2$. Therefore ℓ is even and $\kappa = \pm\lambda^{\ell/2}$. Now $\eta = \pm\lambda^{\ell/2} I$, so the above polynomial f has degree $\ell/2$. But $\ell \neq 0$ so $\ell > \ell/2$ for a contradiction. the result follows. □

We now display an algebra homomorphism $\Delta \to \Lambda$.

Lemma 0.15. *There exists a unique $\mathbb{F}$-algebra homomorphism $\pi : \Delta \to \Lambda$ that sends*

$$A \mapsto q\mathcal{A} + q^{-1}\mathcal{A}^{-1}, \qquad B \mapsto q\mathcal{B} + q^{-1}\mathcal{B}^{-1}, \qquad C \mapsto q\mathcal{C} + q^{-1}\mathcal{C}^{-1}.$$

The homomorphism π sends

$$\alpha \mapsto \nu I, \qquad \beta \mapsto \nu I, \qquad \gamma \mapsto \nu I, \tag{0.9}$$

where $\nu = (q^2 + q^{-2})\mu + \mu^2$.

Proof. Define

$$A^\vee = q\mathcal{A} + q^{-1}\mathcal{A}^{-1}, \qquad B^\vee = q\mathcal{B} + q^{-1}\mathcal{B}^{-1}, \qquad C^\vee = q\mathcal{C} + q^{-1}\mathcal{C}^{-1}.$$

By Lemma 0.8 and Lemma 0.9,

$$(q + q^{-1})A^\vee + \frac{qB^\vee C^\vee - q^{-1}C^\vee B^\vee}{q - q^{-1}} = \nu I, \tag{0.10}$$

$$(q + q^{-1})B^\vee + \frac{qC^\vee A^\vee - q^{-1}A^\vee C^\vee}{q - q^{-1}} = \nu I, \tag{0.11}$$

$$(q + q^{-1})C^\vee + \frac{qA^\vee B^\vee - q^{-1}B^\vee A^\vee}{q - q^{-1}} = \nu I. \tag{0.12}$$

By (0.10)–(0.12) and since νI is central, the elements $A^\vee$, $B^\vee$, $C^\vee$ satisfy the defining relations for Δ from Definition 0.1. Therefore the homomorphism π exists. The homomorphism π is unique since A, B, C generate Δ. Line (0.9) follows from Definition 0.2 and (0.10)–(0.12). $\square$

Lemma 0.16. *For $g \in \mathrm{PSL}_2(\mathbb{Z})$ the following diagram commutes:*

$$\begin{array}{ccc} \Delta & \xrightarrow{\pi} & \Lambda \\ {\scriptstyle g}\downarrow & & \downarrow{\scriptstyle g} \\ \Delta & \xrightarrow{\pi} & \Lambda \end{array}$$

Proof. The elements ρ, σ form a generating set for $\mathrm{PSL}_2(\mathbb{Z})$; without loss we may assume that g is contained in this set. By Theorem 0.5 the action of ρ on Δ cyclically permutes A, B, C. By Lemma 0.12(iii) the action of ρ on Λ cyclically permutes $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$. By Theorem 0.5 the action of σ on Δ swaps A, B and fixes γ. By Lemma 0.12(iv) and the construction, the action of σ on Λ swaps $\mathcal{A}$, $\mathcal{B}$ and fixes I. The diagram commutes by these comments and Lemma 0.15. $\square$

Theorem 0.17. *The action of $\mathrm{PSL}_2(\mathbb{Z})$ on Δ is faithful.*

Proof. Let g denote an element of $\mathrm{PSL}_2(\mathbb{Z})$ that fixes everything in Δ. We show that $g = 1$. By Lemma 0.13 and since ρ, σ generate $\mathrm{PSL}_2(\mathbb{Z})$, there exists an invertible $\xi \in \Lambda$ such that $g(\eta) = \xi\eta\xi^{-1}$ for all $\eta \in \Lambda$. By assumption g fixes the element A of Δ. Under the homomomorphim $\pi : \Delta \to \Lambda$ the image of A is $q\mathcal{A} + q^{-1}\mathcal{A}^{-1}$, so g fixes $q\mathcal{A} + q^{-1}\mathcal{A}^{-1}$ in view of Lemma 0.16. Therefore ξ commutes with $q\mathcal{A} + q^{-1}\mathcal{A}^{-1}$. Recall $\mathcal{A} + \mathcal{A}^{-1} = \mu I$ by Lemma 0.8, so ξ commutes with $\mathcal{A} + \mathcal{A}^{-1}$. By these comments and $q^2 \neq 1$ we find ξ commutes with $\mathcal{A}$. By a similar argument ξ commutes with $\mathcal{B}$. Now $\xi \in Z(\Lambda)$ by Lemma 0.10. Consequently g fixes everything in Λ, so $g = 1$ by Lemma 0.14. $\square$

References

[1] Alperin R.C., Notes: $\mathrm{PSL}_2(\mathbb{Z}) = \mathbb{Z}_2 \star \mathbb{Z}_3$, *Amer. Math. Monthly* **100** (1993), 385–386.